Algebra 1 — Simplifying expressions · Brackets · Solving equations

No calculator. Marks are shown in brackets. ★ = harder question (grades 7–9). Check every solution by substituting it back into the equation.

A · SIMPLIFY AS MUCH AS POSSIBLE

1 $7pq - 2qp$ [1]

> pq and qp are the same term (order of multiplication does not matter): $7pq - 2pq = 5pq$.

2 $3a^2b - ab^2 + 2ba^2 + 4b^2a$ [1]

> Collect like terms: a^2b terms: $3a^2b + 2a^2b = 5a^2b$; ab^2 terms: $-ab^2 + 4ab^2 = 3ab^2$. Answer $5a^2b + 3ab^2$ — the two kinds cannot be combined.

3 $3c^3 \times 4c^2$ [1]

> Multiply the numbers, add the powers: $12c^5$.

4 $2y \times (3y)^2$ [1]

> $(3y)^2 = 9y^2$, so $2y \times 9y^2 = 18y^3$.

5 $7m - (3n - 2m) + n$ [2]

> The minus applies to both terms in the bracket: $7m - 3n + 2m + n = 9m - 2n$.

B · SOLVE FOR x

6 $4 = \dfrac{x}{9}$ [2]

> Multiply both sides by 9: $x = 36$. Check: $36 \div 9 = 4$ ✓

7 $4 = \dfrac{36}{x}$ [2]

> Multiply both sides by x: $4x = 36$, so $x = 9$. Check: $36 \div 9 = 4$ ✓

8 $6(3 - 2x) = 6$ [2]

> Either divide both sides by 6 first: $3 - 2x = 1 \Rightarrow 2x = 2 \Rightarrow x = 1$; or expand: $18 - 12x = 6 \Rightarrow 12x = 12 \Rightarrow x = 1$. Check: $6(3 - 2) = 6$ ✓

9 $5x + 7 = 31 - 3x$ [2]

> Add $3x$ to both sides: $8x + 7 = 31$. Subtract 7: $8x = 24$, so $x = 3$. Check: $15 + 7 = 22$ and $31 - 9 = 22$ ✓

10 $3(x - 1) - (x + 4) = 5$ [2]

> Expand carefully: $3x - 3 - x - 4 = 5 \Rightarrow 2x - 7 = 5 \Rightarrow 2x = 12 \Rightarrow x = 6$. Check: $3 \times 5 - 10 = 5$ ✓

C · PROBLEMS LEADING TO EQUATIONS

11 The sum of three consecutive even numbers is 150. What are the numbers? [3]

> Let the smallest be x; the others are $x + 2$ and $x + 4$. Then $x + (x + 2) + (x + 4) = 150 \Rightarrow 3x + 6 = 150 \Rightarrow 3x = 144 \Rightarrow x = 48$. The numbers are **48, 50, 52**. Check: $48 + 50 + 52 = 150$ ✓

12 Three angles meet on a straight line. Their sizes are $(3x + 10)°$, $2x°$ and $(4x - 10)°$. Find x and the size of each angle. [3]

> Angles on a straight line add up to $180°$: $3x + 10 + 2x + 4x - 10 = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$. The angles are $70°$, $40°$ and $70°$ (sum 180 ✓).

13 An isosceles triangle has two equal sides of length $(2x + 5)$ cm and $(4x - 7)$ cm, and a base of $3x$ cm. Find x and the perimeter of the triangle. [3]

> Equal sides: $2x + 5 = 4x - 7 \Rightarrow 12 = 2x \Rightarrow x = 6$. Sides: 17, 17 and 18 cm, so the perimeter is $17 + 17 + 18 = 52$ cm.

14 The length of a rectangle is 3 cm more than its width. Its perimeter is 42 cm. Find the area of the rectangle. [3]

> Width w, length $w + 3$. Perimeter: $2(w + w + 3) = 42 \Rightarrow 4w + 6 = 42 \Rightarrow w = 9$. So the rectangle is 9×12 and the area is 108 cm².

15 Ana is 4 years older than Ben. In 5 years' time the sum of their ages will be 40. How old is each of them now? [3]

> Ben is b, Ana is $b + 4$. In 5 years: $(b + 5) + (b + 4 + 5) = 40 \Rightarrow 2b + 14 = 40 \Rightarrow b = 13$. **Ben is 13, Ana is 17.** Check: in 5 years $18 + 22 = 40$ ✓

Total: 31 marks

Check: like terms only · a minus outside a bracket changes every sign inside · "do the same to both sides" · substitute your answer back.